

A small Kotlin engineering study

What Codeclew can already prove for an agent, where plain text search remains sufficient, and what is still not established.

5/5 engineering concerns covered compatibility, coordinates, Maven, CFG, manifest	16/16 focused Kotlin checks passed zero failures in the frozen slice	1 bounded RELEASE receipt exact source + K2 declaration identity
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Sample-scoped claim

On five representative concerns in Codeclew's public Kotlin 2.4 worker, every selected mechanism had an exact source fragment and a focused executable check. Sixteen selected tests passed. A preserved warm RELEASE navigation receipt additionally returned the complete decision source, a compiler-resolved declaration identity, exact revision authority and content digests in one bounded response.

This supports a practical claim: Codeclew can give an agent decision-ready, freshness-bound Kotlin evidence instead of only a list of textual matches. It does not establish a population-wide success-rate or token advantage.

1 · Study design

The study was intentionally small. It avoids another expensive multi-agent benchmark and asks whether the released Kotlin contour exposes useful evidence across several common engineering concerns.

Slice	Engineering question	Executable evidence
K1	How is a project/compiler pair admitted or rejected?	4 compatibility-policy tests
K2	Are FIR UTF-16 offsets safe after Cyrillic and emoji?	6 coordinate-normalization tests
K3	Can Maven module selection escape or become ambiguous?	1 reactor-selection test
K4	Does an oversized or malformed control-flow graph fail closed?	3 local-CFG tests
K5	Is semantic input authority canonical and replaceable?	2 manifest-authority tests

Selection and execution

The five concerns were selected manually for engineering relevance before the focused green run. They are not random or held out. The source authority is public commit [6281138](#). The focused Gradle invocation ran warm after compilation and reported 16 tests, 0 skipped, 0 failed. Test reports and source digests were checked locally before this article was built.

Text-search baseline

For the same terms, a deterministic `rg -n` baseline returned 160 matching lines across 11 file-task combinations: K1 5 lines/1 file, K2 21/2, K3 82/2, K4 22/2, and K5 30/4. This is not a speed contest: `rg` is excellent for exact strings. The comparison asks what the result means. Text search supplies lexical locations; Codeclew can additionally bind a selected declaration, compilation, exact source window and evidence digest.

2 · What the Kotlin evidence looks like

Compatibility decisions are typed

```
val row = QUALIFIED_KOTLIN_ENGINE_ROWS.singleOrNull {
    it.projectCompilerVersion == project.projectCompilerVersion &&
    it.engineCompilerVersion == engine.analyzerCompilerVersion
} ?: return KotlinEngineCompatibilityDecision(
    status = "REJECTED",
    kind = "UNQUALIFIED",
    reason = "PROJECT_ENGINE_ROW_NOT_QUALIFIED",
    btaEligible = false,
)
```

Nested positions share one byte domain

```
val byteRange = coordinates.range(start, end) ?: return null
normalized += rebuild(argument, mapOf(
    "argumentStart" to JsonPrimitive(byteRange.first),
    "argumentEnd" to JsonPrimitive(byteRange.last + 1),
))
```

Oversized CFGs become boundaries

```
if (rawNodes.size > 4_096 || rawEdges.size > 8_192) {
    return localCfgBoundary(
        "LOCAL_CFG_BUDGET_EXCEEDED", ...
    )
}
```

These excerpts are not free-floating screenshots. The public evidence record links each fragment to a repository revision, file, line range and SHA-256 source digest. On the interactive documentation page, selecting a fact changes the exact code block shown beside it.

3 · Results and interpretation

Observation	Result	Interpretation
Focused correctness	16/16 passed	All selected mechanisms have executable support at the frozen revision.
Concern coverage	5/5	The sample spans policy, coordinates, build topology, bounded graphs and hashing.
Warm RELEASE navigation	1 bounded response	Returned 3 ranked cards, one exact 115-line source window, K2 identity and evidence digest.
Plain text baseline	160 matching lines	Useful lexical discovery, but no compiler identity or freshness authority.
Prior paired agent task	3.795× fewer effective tokens	Both answers were source-correct; this remains one prepared warm task.

Where Codeclew adds value

The advantage is strongest when an agent must choose among similarly named declarations, explain a branch with its exact qualifiers, or carry evidence into a later change. A bounded card can contain the selected declaration, a source window, compilation authority, certainty, limits and a stable digest. That is more decision-ready than a long list of textual hits and safer than an uncited summary.

Where `rg` remains the right tool

For an exact string in a known file, `rg` is simpler and faster. Codeclew should not replace it. The product claim is narrower: Codeclew is useful when navigation requires identity, context, boundedness and evidence that survives review.

4 · Limitations and next evidence

One public module

The five concerns come from Codeclew's Kotlin worker, not five independent repositories.

Manual selection

The sample demonstrates breadth but cannot estimate population performance.

Warm focused run

Cold compilation and installation were excluded; the 784 ms result is not a cold-start claim.

One navigation receipt

Only the preserved compatibility-policy query is used as direct RELEASE navigation evidence.

Fresh-run operational blocker

A new attempt on 1 September stopped before facts with RESOURCE_LIMIT because the long-lived local CAS catalog snapshot exceeded its bound. The failed attempt is not counted as a successful case.

Prior token result remains descriptive

The 3.795× token result and 40.28× tool-output result come from one prepared warm paired task and do not prove general superiority.

Minimal next gate

For a stronger public claim, freeze six small tasks across three real Kotlin repositories: two location/explanation tasks per repository, balanced Default/Codeclew order, identical model and prompt budget, warm state on both arms, and a blinded source-correctness check. Stop after six tasks. That is enough to test whether the observed advantage transfers without rebuilding an industrial benchmark harness.

5 · Evidence ledger

Source revision	6281138ecbf73bc5de1a9c7eae2cdf7009e6ca1
Kotlin profile	kotlin-2.4.10-gradle-single
Compilation	:workers:kotlin/main
Focused test result	16 tests / 0 failures / 0 skipped
Worker.kt SHA-256	b4a5bdd058e9d400e2955dbc8115893b0911909fc446abc5a08c5c86992d4e44
MavenProjectModel.kt SHA-256	9c5ae9638c36e7072d50cce1f2518f5672e42a4075dda2d543404310cf6b386e
LocalCfgIndex.kt SHA-256	07dd66f68183c25df4d56b3b814769bdd8f898a490d3e368e2e8f7a1fd8a06e3
SemanticInputManifestAuthority.kt SHA-256	8dbffc93e43a5ba236f91229f99c7343ff64cfdac4c270f1c5150c98529d7ea4
Preserved navigation receipt SHA-256	ea21155aa47f533394a64e67b6f8dda29e1b0f98d27ddf29e6af24414e4b85b4
Paired Q1 evidence receipt SHA-256	61ec5bcc2583b09e9d1c6d4539055e46031762118b62bbe87de4aa68da808ab2

Conclusion

The small study justifies using Codeclew today for bounded Kotlin evidence navigation in prepared projects. It does not justify removing text search, claiming universal token savings, or ignoring the catalog resource-limit blocker. The useful product direction is clear: keep the response compact and evidence-bound, make accumulated-state maintenance reliable, and validate transfer on a six-task, three-repository follow-up.